

GOLDEN GATE UNIVERSITY

Doctor of Business Administration (Generative AI)

ASD Literature Review

Consolidated Standalone Document

Companion to v3 Dissertation (E38 + E40) + Main Thesis Ch. 2 (424-entry bibliography)

A Governance Framework for Responsible Explainable AI-Assisted Retrospective EEG-Based Neuropsychiatric Decision Support Under Human Clinical Oversight

*Integrating Generative AI, Retrieval-Augmented Generation,
and Continuous Monitoring for Clinical and Consumer Healthcare Applications*

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This consolidated ASD literature review unifies: (1) the March-2026 Chapter 2 Literature Review Autism ASD document; (2) v3 dissertation E38 substantive lit review excerpt + E40 flow-document anchors; (3) the candidate's own 2025 ICSIT publication; (4) PRISMA cascade (2,847 papers → 156 included) across 6 thematic areas; aligned with main.pdf B2C-primary framing and the dissertation v3 GGU-strict structure.

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1. Chapter Claim + Coverage Scope

The chapter claim is the single load-bearing sentence the dissertation defends; coverage scope is the PRISMA cascade output.

Chapter claim. *The existing ASD-EEG + AI-governance literature demonstrates high single-axis classification accuracy (78–99% across 50+ reviewed methods) but reveals five critical gaps (theoretical + practical + technical + governance + methodological) that collectively prevent unified, explainable, and governance-embedded clinical AI deployment for pediatric ASD screening support under clinician oversight.*

Cascade stage	Count	Action
Database query (2015–2025)	2,847	IEEE + PubMed + Scopus + WoS; ASD-EEG + AI governance + XAI + HITL keywords.
Title + abstract screen (PRISMA)	612	Inclusion: ASD-EEG OR AI governance OR XAI OR pediatric AI.
Full-text quality review	156	Quality: peer-reviewed + reproducible + reported metrics.
Theme coding	6 themes	Per-theme tables (T1–T6) and per-theme synthesis.

Table 1: PRISMA cascade 2,847 → 156 papers across 6 thematic areas.

2. Theme 1 — Classical EEG and Machine Learning Frameworks

Classical EEG analysis spans spectral coherence, wavelet, entropy, and statistical features fed to SVM / RF / KNN / LDA / XGBoost classifiers. The accuracy range 70–90% sets the baseline that RGAIG-wrapped methods must match or exceed while adding governance.

Author / Year	Method	Accuracy	Contribution + relevance
Duffy & Als (2012) [49]	FFT + PCA + DFA	—	$n=1,304$ large-scale EEG; foundational ASD-vs-neurotypical spectral coherence work.
Sinha et al. (2022) [38]	DWT + KNN + Shannon entropy	94.60%	Efficient classical-ML approach; IETE Journal of Research.
Alturki et al. (2021) [34]	CSP + LBP + KNN	98.46%	Common Spatial Pattern; spatial filtering excellence.
Pham et al. (2020) [36]	HOS bis-spectrum + PNN	98.70%	Higher-order spectra; non-linear phase coupling.
Haputhanthri et al. (2019)	RF + correlation FS	93.00%	Low-cost system; minimal EEG channels.
Harun et al. (2018)	ANN + SVM emotion-based	—	ASD diagnosis via emotional EEG response patterns.

Table 2: Theme 1 — Classical ML for ASD-EEG: 6 representative works, accuracy range 70–99%.

3. Theme 2 — Deep Learning-Based EEG Frameworks

Deep learning architectures (BiLSTM, CNN, EEGNet, Transformer, ViT) achieve 78–99% accuracy with the trade-off of reduced interpretability. The dissertation's two-layer XAI (caregiver + clinician) addresses this trade-off.

Author / Year	Architecture	Accuracy	Innovation + relevance
Asthana et al. (2025) [31]	CNN-ASD + CWT hybrid	78.35%	Candidate's own published work ; adaptive filter bank for non-stationary EEG; 10-fold CV; outperforms BiLSTM-Attention + ResNet50.
Chen et al. (2021) [41]	BiLSTM + attention	—	Temporal-dependency capture + interpretability gain.
Te et al. (2021) [42]	Spectrogram + CNN	99.15%	Time-frequency CNN; highest reported DL accuracy ; PLOS ONE.
Lalawat et al. (2024) [32]	ASD-Net + DenseNet + ResNet + TFD	97.35%	SPWVD time-frequency distributions; IEEE Sensors Journal.
Mohi ud Din et al. (2022) [33]	CWT + SqueezeNet	82.98%	CWT scalogram + lightweight CNN.
Singh et al. (2023)	CSTO-optimised deep ResNet	88.60%	Chronological Sewing Training Optimisation.
Zhu et al. (2024)	CNN + LSTM hybrid	74.55%	Spatial + temporal joint learning.
Khan et al. (2023)	ResNet-50 + multiple DL	81.00%	Deep CNN benchmark suite.
Omar et al. (2024)	Multi-input 1D CNN	77–79%	Channel-wise spatial-temporal representation.

Table 3: Theme 2 — Deep Learning for ASD-EEG: 9 architectures, accuracy range 74–99%.

4. Theme 3 — Connectivity and Multimodal Integration

Connectivity-based + multimodal frameworks deepen ASD brain-dynamics understanding via graph-theoretic measures, coherence, phase-locking, fMRI / DTI fusion, and eye-tracking integration.

Author / Year	Method	Accuracy	Contribution + relevance
Barttfeld et al. (2010) [50]	Connectivity	—	Foundational under-connectivity hypothesis: reduced long-range, increased local.
Tan et al. (2023) [37]	Eye-tracking + EEG	—	Multimodal neurocognitive biomarkers; IEEE TCDS.
Wadhera et al. (2023) [35]	Graph-theoretic + SVM	92.34%	Network metrics for social cognition analysis.
Ke et al. (2024) [48]	Retrospective + spectral + connectivity	high	$n=288$ cohort; connectivity as major diagnostic marker.
Xu et al. (2024)	Coherence + PLV	—	Robustness of phase-locking-value connectivity.
Cociu et al. (2018) [47]	EEG + fMRI + DTI fusion	—	Multimodal network abnormality complementarity.
Lalawat et al. (2025)	fMRI + EEG deep learning	—	Multimodal schizophrenia framework with ASD applications.

Table 4: Theme 3 — Connectivity + Multimodal Integration: 7 representative works.

5. Theme 4 — Feature Optimisation and Signal Decomposition

Feature-optimisation + decomposition methods (VMD, SWT, CSP, channel selection) refine ASD detection by extracting more discriminative or computationally efficient features.

Author / Year	Optimisation method	Contribution
Lalawat et al. (2025) [40]	Optimal VMD for EEG stress recognition	Decomposition adaptability to neurological-disorder analysis; cross-domain transfer.
Fahmi et al. (2025) [46]	SWT + FLDA	Stationary Wavelet + Fisher LDA; clear separability ASD vs control.
Alturki et al. (2021) [34]	Common Spatial Pattern (CSP)	98.46% accuracy; differentiates ASD + epilepsy.
Aslam et al. (2022)	Channel selection (DWT + CSP + entropy + correlation)	Reviews computational efficiency vs. interpretability trade-offs.

Table 5: Theme 4 — Feature Optimisation + Decomposition methods.

6. Theme 5 — Early Detection and Developmental Insights

Early detection (infant + preschool cohorts) is the highest-impact use-case for ASD screening: earlier intervention = better outcomes.

Author / Year	Cohort	Accuracy	Significance
Bosl et al. (2018) [43]	Infants	—	Nonlinear EEG metrics predict ASD risk before behavioural symptoms appear.
Peck et al. (2021) [44]	Infants	100.00%	EEG response to speech; ultra-early prediction.
Santarone et al. (2023) [39]	Preschool (n=cohort)	78% abnormality	EEG features abnormal in 78% of preschool ASD children.
Esqueda-Elizondo et al. (2022) [45]	13-year-old (case study)	—	Theta/Beta + MLP for ASD attention assessment.

Table 6: Theme 5 — Early Detection and Developmental Insights.

7. Theme 6 — Governance + Trust + Adoption (RGAIG-Specific)

The governance + trust + adoption literature has been single-axis (TAM, TOE, RBV, NIST, ISO, EU AI Act); no prior work integrates these axes into an audit-bearing fabric for pediatric ASD screening. RGAIG is the integrating contribution.

Cite	Construct / standard	Contribution + role in RGAIG
NIST AI RMF (2023) [11]	4-function framework	Govern + Map + Measure + Manage; US anchor.
ISO/IEC 42001 (2023) [12]	AI Management System	First certifiable; PDCA + Annex A.
EU AI Act (2024) [14]	Risk-tier + Art. 13/14/86	Transparency + oversight + right-to-explanation.
DPDP Act (2023) [15]	India data protection	Pediatric DPDP-compliant pathway.
Mittelstadt et al. (2019) [1]	Principles + critique	“Principles alone” — operationalise via RGAIG controls.
Davis (1989) [3]	TAM	Extended with mediator (M1, M2) + moderator (W) in RGAIG.
Mayer et al. (1995) [4]	Trust integrative model	3-dim trust (ability + benevolence + integrity).
Lee & See (2004) [5]	Trust in automation	Calibration framework; 5-phase trust journey.
Lundberg & Lee (2017) [6]	SHAP	Game-theoretic XAI for clinician layer.
Reddy et al. (2020) [7]	AI governance healthcare	Identifies governance gap; RGAIG closes it.

Table 7: Theme 6 — Governance + Trust + Adoption (10 anchor works).

8. Critical Synthesis Across Six Themes

The synthesis below names what the literature collectively offers and what RGAIG closes.

Across the six themes, the ASD-EEG + AI-governance literature offers mature single-axis frameworks for neurophysiological biomarkers (delta-gamma + connectivity + ERPs), classical ML (70–99% accuracy via SVM / RF / XGBoost / CSP / HOS), deep learning (74–99% via 1D-CNN / 2D-CNN / EEG-Net / BiLSTM / Transformer), connectivity + multimodal (graph-theoretic + EEG-fMRI-DTI fusion + eye-tracking), feature optimisation (VMD / SWT / CSP / channel selection), early detection (infant + preschool cohorts), and governance + trust + adoption (NIST + ISO + EU AI Act + DPDP + TAM + Trust). However, no prior work integrates these axes into a single, operationalisable, audit-bearing governance fabric for B2C-primary, clinician-supported pediatric ASD screening support. The proposed RGAIG framework closes this integration gap (G1–G5) through a 5-layer + transverse-audit architecture, anchored by 24 controls + 6-level maturity instrument + multi-jurisdiction crosswalk + the candidate’s own 2025 ICSIT-published CNN-ASD baseline [31] as the methodological starting point.

9. Five Research Gaps G1–G5 (Closure Commitment)

The literature reveals 5 typed gaps; RGAIG closes all 5 with named mechanisms.

Gap	Type	RGAIG closure mechanism
G1	Theoretical	RGAIG construct integrating RAI + governance + trust + adoption (5 layers + transverse audit).
G2	Practical	RGAIG 6-level maturity instrument (operational + scored + auditable).
G3	Technical	7-layer reference architecture: data + AI + governance + HITL + value + audit + transverse monitoring.
G4	Governance	Multi-jurisdiction crosswalk: NIST AI RMF + ISO/IEC 42001 + EU AI Act + DPDP + GDPR + HIPAA.
G5	Methodological	Pragmatist mixed-methods DSR + PLS-SEM + qualitative expert panel + retrospective benchmark hybrid.

Table 8: Five research gaps G1–G5 with RGAIG closure mechanism.

10. Consolidated Reference List (50 Citations)

The 50 citations below are drawn from the v3 dissertation References table; full main thesis bibliography lives in `references.bib` (424 entries). This list provides the load-bearing 50 references that anchor every theme.

Cite	Reference (short)
[1]	Mittelstadt et al. (2019) — <i>Principles alone cannot guarantee ethical AI</i>
[2]	Habli et al. (2020) — <i>AI in healthcare: assurance + governance</i>
[3]	Davis (1989) — <i>Technology Acceptance Model (TAM)</i>
[4]	Mayer, Davis & Schoorman (1995) — <i>Trust integrative model</i>
[5]	Lee & See (2004) — <i>Trust in automation</i>
[6]	Lundberg & Lee (2017) — <i>SHAP attribution</i>
[7]	Reddy et al. (2020) — <i>AI governance in healthcare</i>
[8]	Djermal et al. (2017) — <i>KAU ASD-EEG dataset</i>
[9]	Di Martino et al. (2017) — <i>ABIDE-II dataset</i>
[10]	Lord et al. (2020) — <i>ASD diagnostic instruments (ADOS / ADI-R)</i>
[11]	NIST (2023) — <i>AI RMF 1.0</i>
[12]	ISO/IEC 42001 (2023) — <i>AI Management System Standard</i>
[13]	Khare et al. (2023) — <i>ASD EEG classification benchmarks</i>
[14]	EU AI Act (2024) — <i>EU Regulation</i>
[15]	DPDP Act (2023) — <i>India Data Protection</i>
[16]	Ribeiro et al. (2016) — <i>LIME</i>
[17]	Wachter et al. (2017) — <i>Counterfactual explanations under GDPR</i>
[18]	Sundararajan et al. (2017) — <i>Integrated Gradients</i>
[19]	Doshi-Velez & Kim (2017) — <i>Interpretability rigour</i>
[20]	Friedler et al. (2019) — <i>Comparative fairness study</i>
[21]	Hardt et al. (2016) — <i>Equality of opportunity</i>
[22]	Selbst et al. (2019) — <i>Fairness + ML beyond model property</i>
[23]	Holstein et al. (2019) — <i>Improving fairness in practice</i>
[24]	Lewis et al. (2020) — <i>Retrieval-Augmented Generation (RAG)</i>
[25]	Gao et al. (2023) — <i>RAG survey: naive → agentic</i>
[26]	Shi et al. (2024) — <i>Hallucination taxonomy</i>
[27]	Anthropic (2024) — <i>Model Context Protocol (MCP)</i>
[28]	Tornatzky & Fleischer (1990) — <i>TOE framework</i>
[29]	Barney (1991) — <i>Resource-Based View (RBV)</i>
[30]	Cidav et al. (2017) — <i>Autism economic burden</i>
[31]	Asthana, Lalawat, Baudh & Anand (2025) — CNN+CWT ASD-EEG diagnosis (2025 ICSIT) — candidate's own published work
[32]	Lalawat et al. (2024) — <i>TFD-based ASD detection (IEEE Sensors)</i>
[33]	Mohi ud Din et al. (2022) — <i>CWT + CNN ASD classification</i>
[34]	Alturki et al. (2021) — <i>CSP + EEG for ASD + epilepsy</i>
[35]	Wadhwa & Kakkar (2021) — <i>Graph-theoretic EEG</i>
[36]	Pham et al. (2020) — <i>HOS bispectrum + EEG</i>
[37]	Tan et al. (2022) — <i>Eye-tracking + EEG</i>
[38]	Sinha et al. (2022) — <i>Efficient ASD detection via EEG</i>
[39]	Santarone et al. (2023) — <i>Preschool ASD EEG features</i>
[40]	Lalawat et al. (2025) — <i>VMD stress classification (Applied Acoustics)</i>
[41]	Chen et al. (2021) — <i>BiLSTM + attention for ASD</i>
[42]	Te et al. (2021) — <i>Spectrogram CNN 99.15% (PLOS ONE)</i>
[43]	Bosl et al. (2018) — <i>Infant EEG nonlinear metrics</i>
[44]	Peck et al. (2021) — <i>Infant EEG + speech 100%</i>
[45]	Esqueda-Elizondo et al. (2022) — <i>Theta/Beta + MLP</i>
[46]	Fahmi et al. (2025) — <i>SWT-FLDA</i>
[47]	Cociu et al. (2018) — <i>EEG + fMRI + DTI fusion</i>
[48]	Ke et al. (2024) — <i>n=288 spectral + connectivity</i>
[49]	Duffy & Als (2012) — <i>n=1304 spectral coherence + DFA</i>
[50]	Barttfeld et al. (2010) — <i>Under-connectivity hypothesis</i>

Table 9: Consolidated 50-reference list aligned with v3 dissertation + main thesis bibliography.

End of Consolidated ASD Literature Review. 6 themes; 50 citations; 5 gaps closed by RGAIG; PRISMA cascade documented. Aligned with main.pdf (1127 pp, 424-entry references.bib), v3 dissertation (E38 +

E40), and the candidate's own 2025 ICSIT publication. Companion to: Ch2_Flow_Document_ASD.pdf, asd_questionnaire.pdf, and the full main thesis Ch. 2.